

UNIVERSITÀ DEGLI STUDI DI MILANO

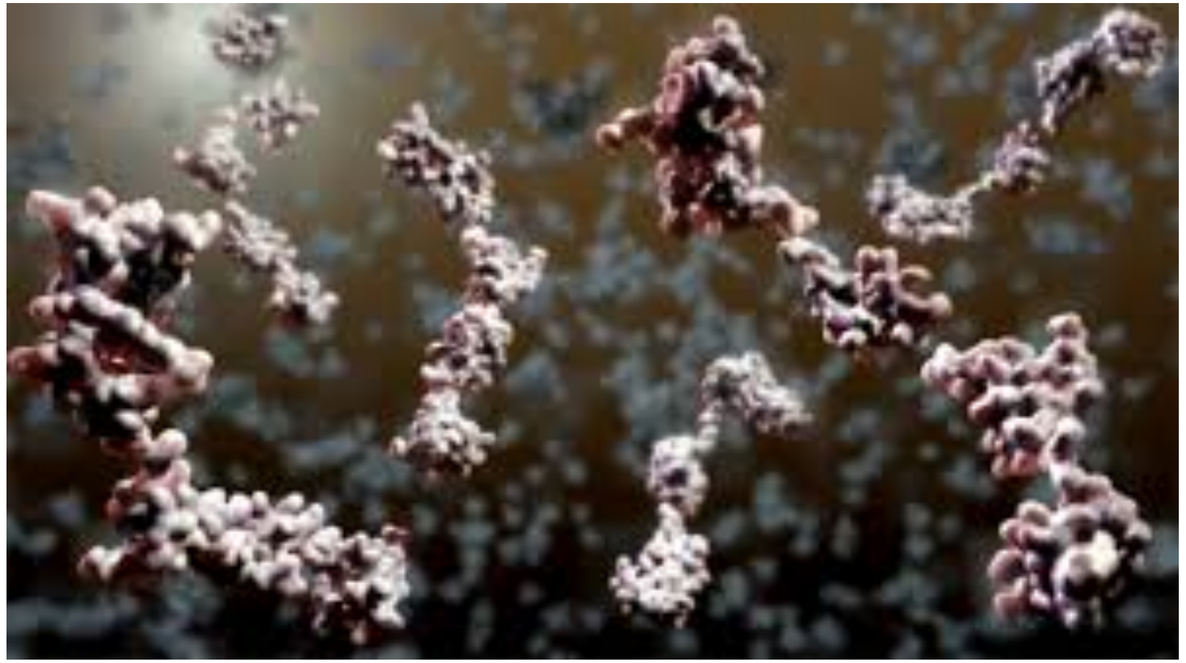
A mechano-statistical perspective of biomolecules in motion: towards building an *in-silico* microscope

Carlo Camilloni

Why a Statistical and Mechanical perspective

When we observe/describe biomolecules we generally handle many molecules, made of many atoms, that can:

- change their shape;
- move in 3D space;
- have interactions;



To describe all these we need to know how they move and interact (mechanics) and how to handle large numbers (statistics)

Why a Statistical and Mechanical perspective

In the first part we will see how statistical observations can be interpreted using the language of mechanics: energy.

In the second part we will revisit classical experiments on protein folding and ligand binding (kinetics, calorimetry and single-molecule methods) that build on this connection between probabilities, energies and rates.

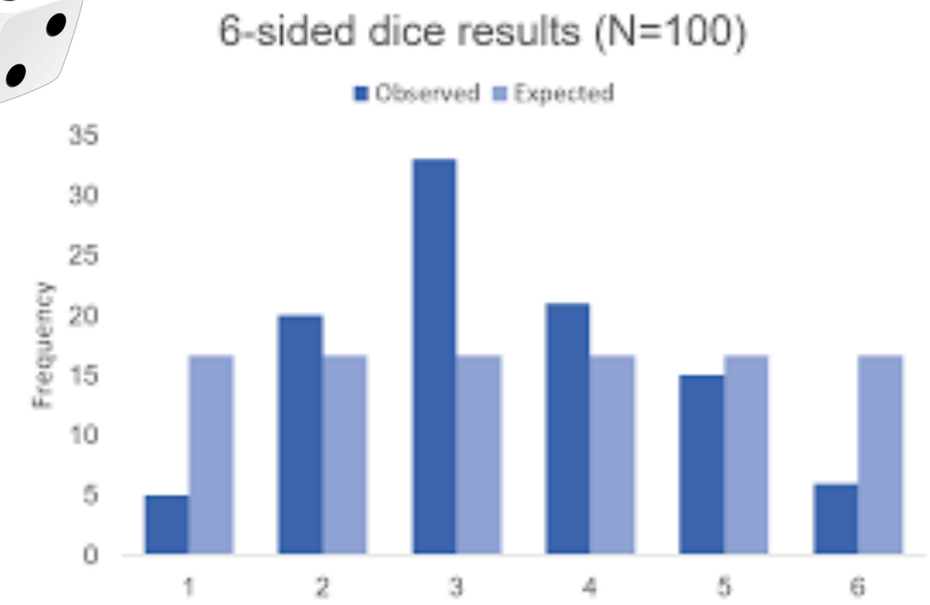


Probabilities are estimated using histograms: the problem of sampling relative and absolute probabilities



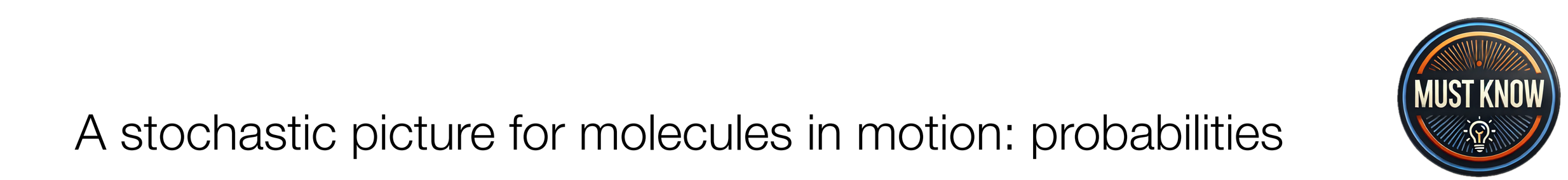
Stochastic events are estimated by SAMPLING. Sampling means exploring a probability distribution looking at many outcomes organised as histograms.

Absolute probability needs to have information on all outcomes, relative probabilities are just frequency ratios:



Absolute probability: p(i) = count(i)/(sum_k(count(k)))
Relative probability: p(j)/p(i) = count(j)/count(i)

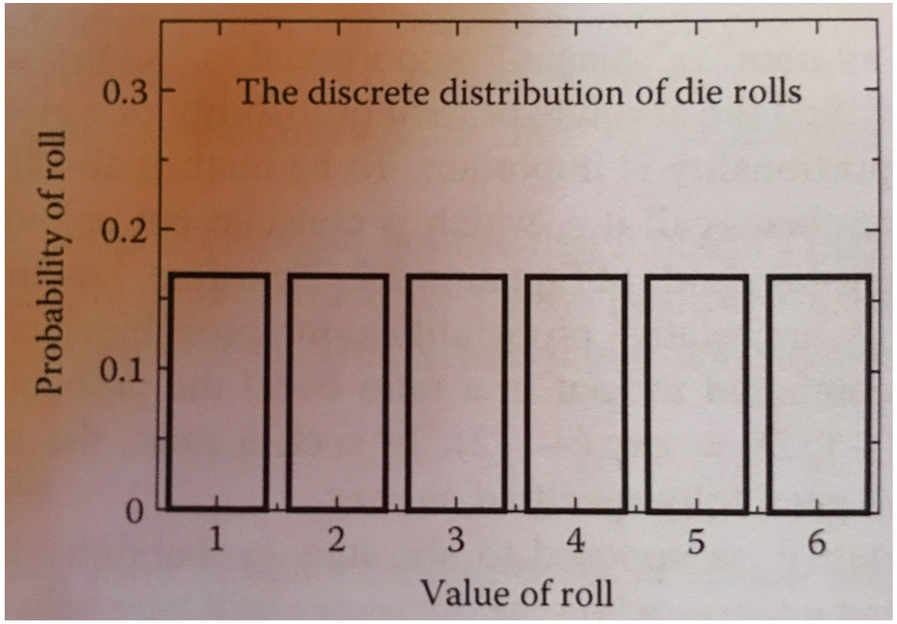
A stochastic picture for molecules in motion: probabilities



Stochastic means randomly determined, events happen by chance, by observing many events one can learn **what is the probability of the different outcomes** that is one can learn their **probability distribution**.

If some outcome is more likely than another one could think in terms of energy that that outcome is favourable in **energy**.





p(j)=1/6 with j=1 to 6 **Discrete**

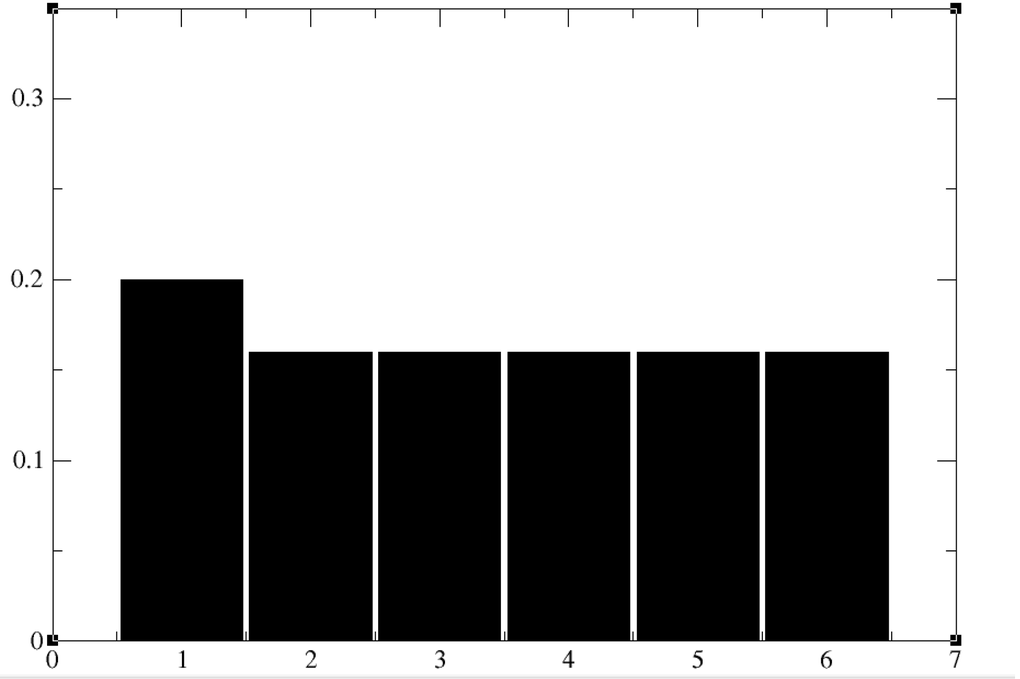
$$\sum_{j=1}^{6} p(j) = 1$$

Normalised
When all possibilities are accounted for the probability is 100%

A stochastic picture for molecules in motion: biases

If your die is biased towards a given number, let's say 1, then your histogram will be higher at 1. In terms of energy you can think that the state 1 is energetically more favourable.





$$\sum_{j=1}^{6} p(j) = 1$$

Normalised
When all possibilities are accounted for the probability is 100%

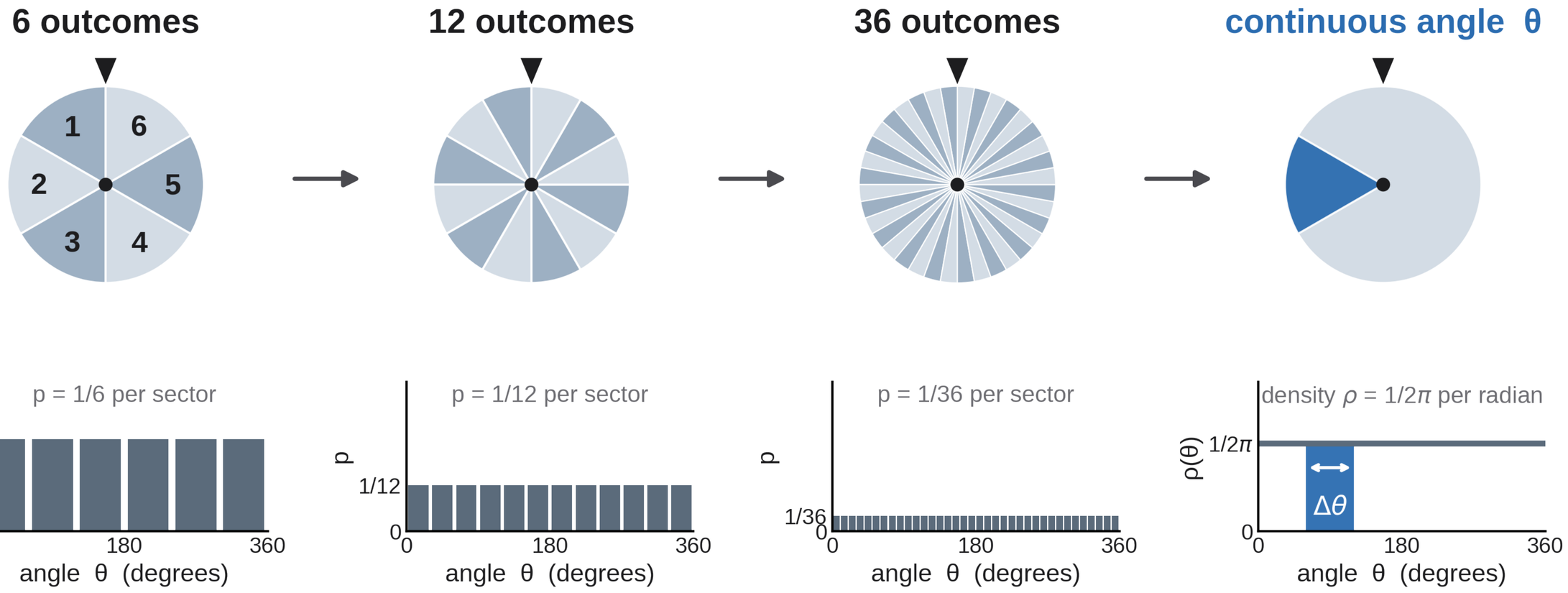
To bias a die you change its mechanics (e.g. its centre of mass) in the presence of a force (e.g. gravity).

Summary 1

Like a die, a molecule obeys deterministic mechanics, yet the outcome is random: it depends on details we cannot control.
In water, each atom is kicked by solvent molecules ~10^{13} times per second; these thermal kicks (energy ~kT) erase the memory of the starting point.
So each observation is a random draw: outcomes within a few kT are all seen, outcomes many kT higher almost never.



Stochastic processes may have a continuous probability distribution

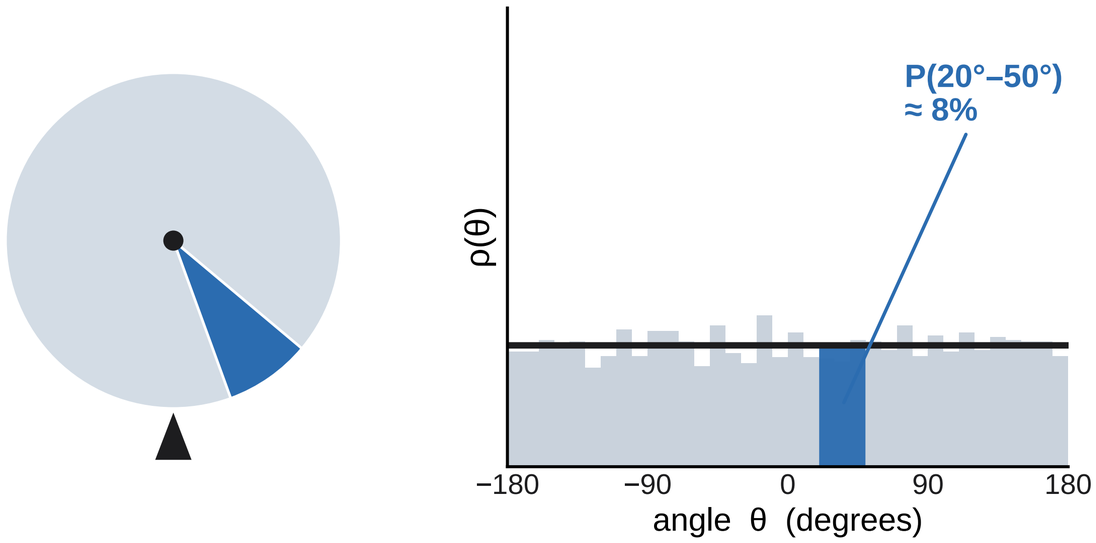


smaller sectors: each p shrinks, but p / Δθ stays constant

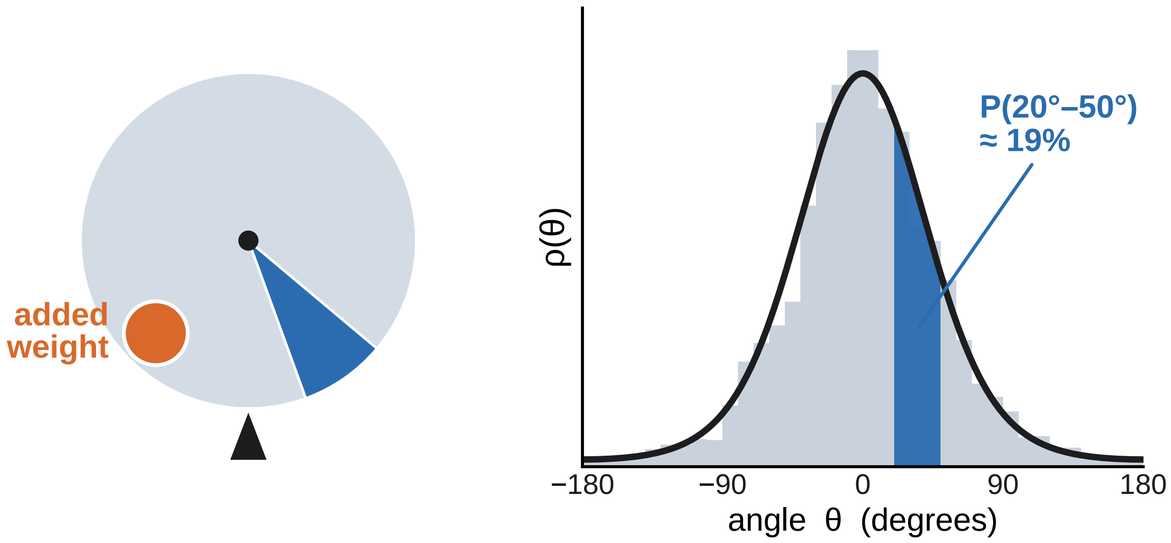
$P(a < \theta < b) = \rho\,\Delta\theta = 1/6$ **(one die face)**

Stochastic processes may have a continuous probability distribution

Fair wheel



Wheel with an added weight



same rule for both: $P(a<\theta<b) = \int \rho \, d\theta$

grey: histogram of 3000 spins · black: density ρ(θ)

Densities have units of 1/x

$$\rho_G(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-<x>)^2}{2\sigma^2}}$$

Gaussian

Bulk observables: Average and Standard Deviation



Experiments rarely show single outcomes, only bulk or state averages.

In practice we **estimate** averages using sums:

$$< f > \doteq \frac{1}{N} \sum_{i=1}^{N} f(x_i)$$

Variance and Standard deviation:

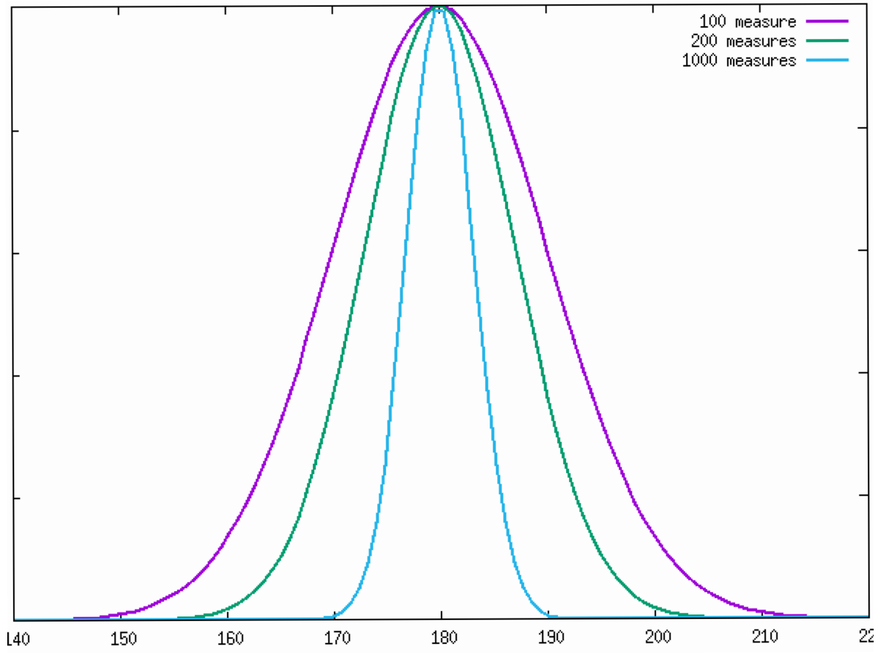
$$\sigma^2 = var(f) = < (f - < f >)^2 > = \int dx (f(x) - < f >) \rho(x) \doteq \frac{1}{N-1} \sum_{i=1}^{N} (f(x_i) - < f >)^2$$

Tells us about the width of the distribution ρ(x).

Standard Error of the Mean

Standard error: $\text{Std-err} = \sqrt{\frac{var(f)}{N}} = \frac{\sigma}{\sqrt{N}} = \sqrt{\frac{1}{N^2 - N}\sum_{i=1}^{N}(f(x_i) - < f >)^2}$

Tells us about the **precision of our estimate of the average,** the idea is that our average value that we estimate from multiple observations has a resolution with the shape of a Gaussian with the width of the standard error.

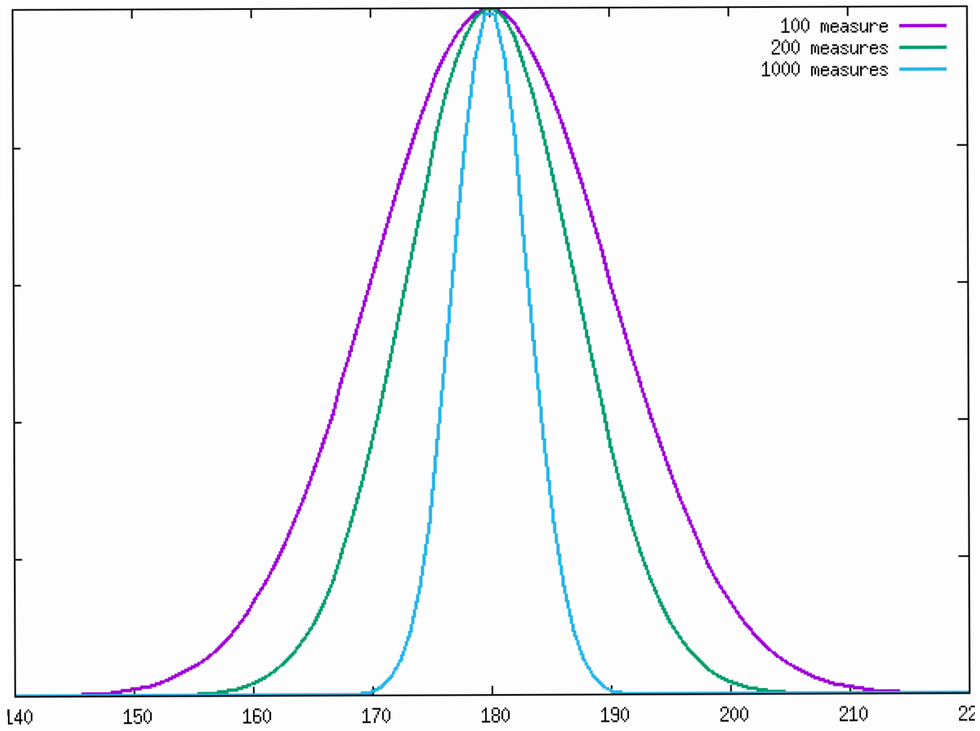


Average, Standard Deviation, Standard Error of the Mean

$$< f > \doteq \frac{1}{N}\sum_{i=1}^{N} f(x_i) \qquad \sigma^2 = var(f) = < (f - < f >)^2 > = \int dx (f(x) - < f >)\rho(x) \doteq \frac{1}{N-1}\sum_{i=1}^{N}(f(x_i) - < f >)^2$$

$$\text{Std-err} = \sqrt{\frac{var(f)}{N}} = \frac{\sigma}{\sqrt{N}} = \sqrt{\frac{1}{N^2 - N}\sum_{i=1}^{N}(f(x_i) - < f >)^2}$$

The average of a random process is (for large N) distributed as a Gaussian of width std-err and so you can set confidence intervals: (±1std-err is 68%, ±2std-err 95%, ±3std-err 99.7%, …). This is the central limit theorem.



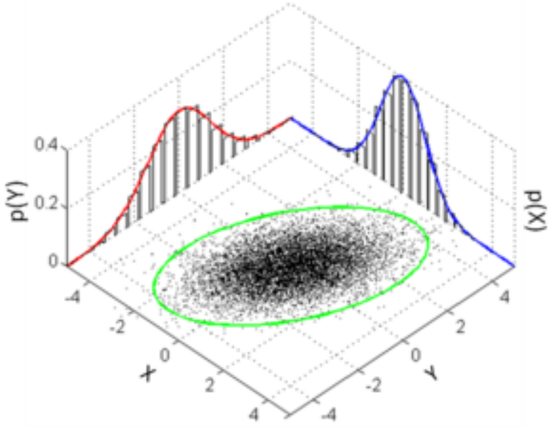
Multidimensional distribution function: projection and correlation

Probability distributions can be more than 1D. For example the probability distribution of a molecule around a receptor may be studied in terms of (x,y,z) coordinates of the center-of-mass, in this case would be 3D.

$$\int \rho(x,y)dxdy = 1$$ Normalisation

How can we show a multi dimensional distribution in less dimen

$$\rho(x) = \int \rho(x,y)dy$$ Projection/Marginalisation



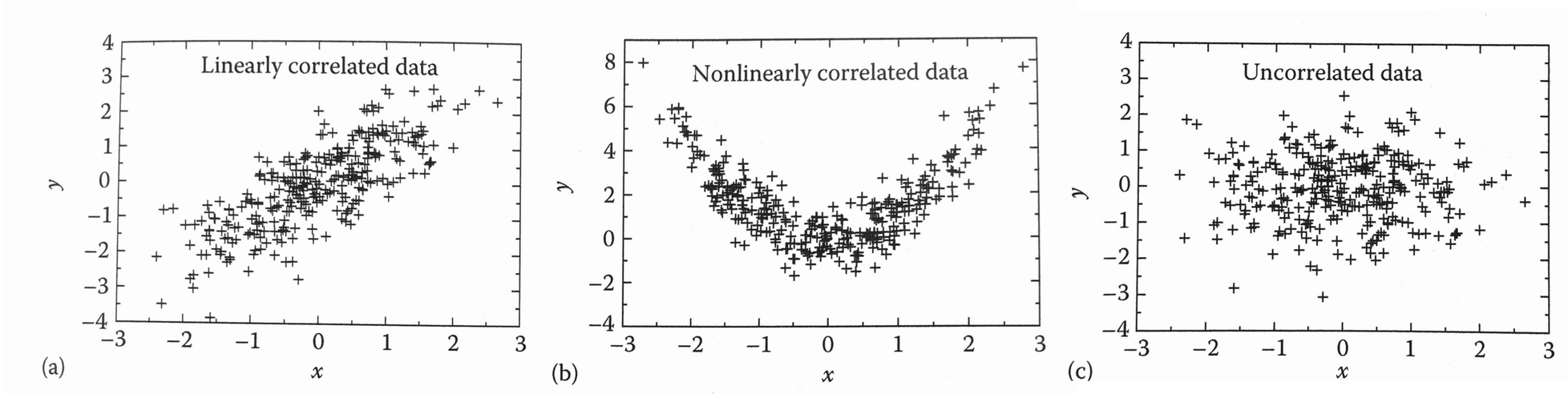
$$< x >= \int dxdyx\rho(x,y)$$

$$< y >= \int dxdyy\rho(x,y)$$

$$< xy > \int dxdyxy\rho(x,y)$$

By looking at the data can you think of a projection that is more important to understand the data?

Multidimensional distribution function: correlation



$$< x >= \int dxdyx\rho(x,y)$$

$$< y >= \int dxdyy\rho(x,y)$$

$$< xy > \int dxdyxy\rho(x,y)$$

If two variables are independent then

$$\rho(x,y) = \rho(x)\rho(y) \quad \textbf{So}$$

$$< xy >=< x >< y >$$

So we have correlation if

$$< xy > - < x >< y >\neq 0$$

Pearson correlation coefficient:

$$r = \frac{< xy > - < x >< y >}{\sigma_x \sigma_y}$$

Statistical Mechanics: the Boltzmann distribution



What is the connection between statistics and mechanics?

That is the Boltzmann equation?

$$pdf(q,v) = \rho(x,v) \propto \exp\left[\frac{-E(x,v)}{k_B T}\right] = \exp\left[\frac{-U(x)}{k_B T}\right] \exp\left[\frac{-K(v)}{k_B T}\right]$$

is the equilibrium distribution of a system in thermal contact with a bath at temperature T

$$pdf(v) \equiv \rho(v) \propto \exp\left[\frac{-K(v)}{k_B T}\right] = \exp\left[\frac{-\frac{1}{2}mv^2}{k_B T}\right]$$

$$pdf(x) \equiv \rho(x) \propto \exp\left[\frac{-U(x)}{k_B T}\right]$$

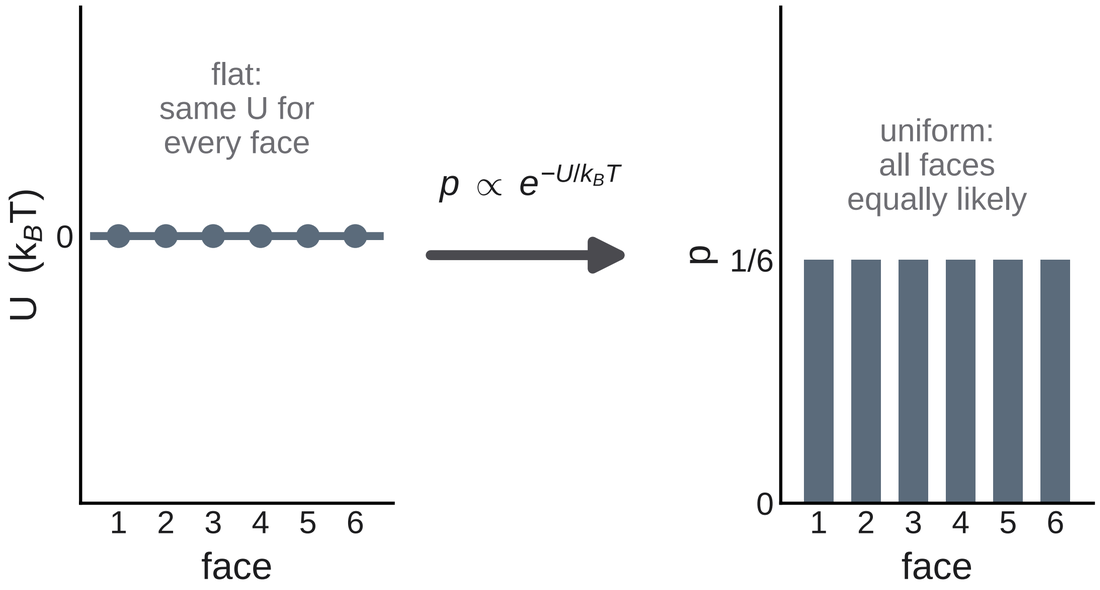
This gives a link between the energy of a conformation and the probability of observing it

Statistical Mechanics: the Boltzmann distribution

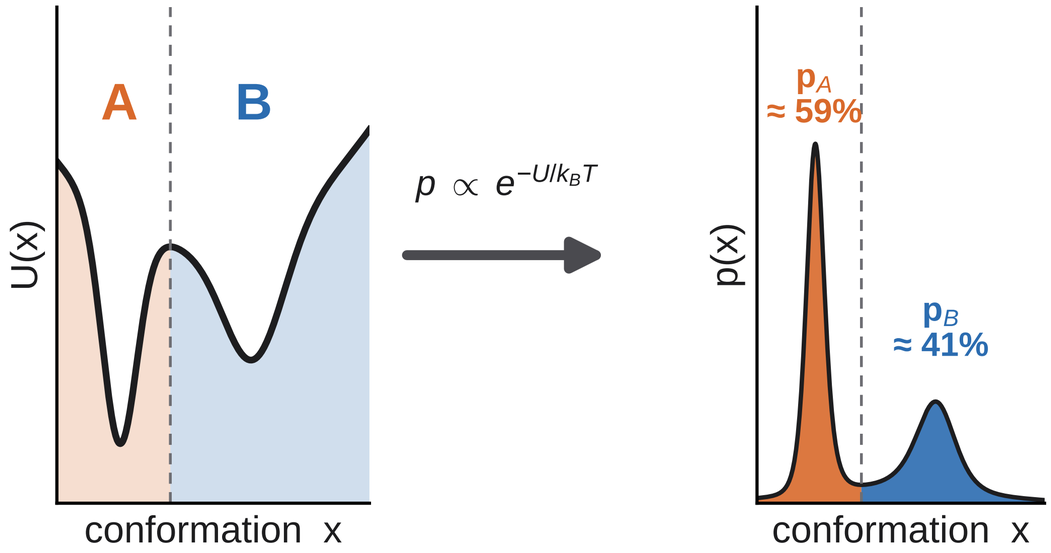
$$pdf(x) \equiv \rho(x) \propto \exp\left[\frac{-U(x)}{k_B T}\right]$$

This gives a link between the energy of a conformation and the probability of observing it

Dice: flat potential energy



Two basins: A and B



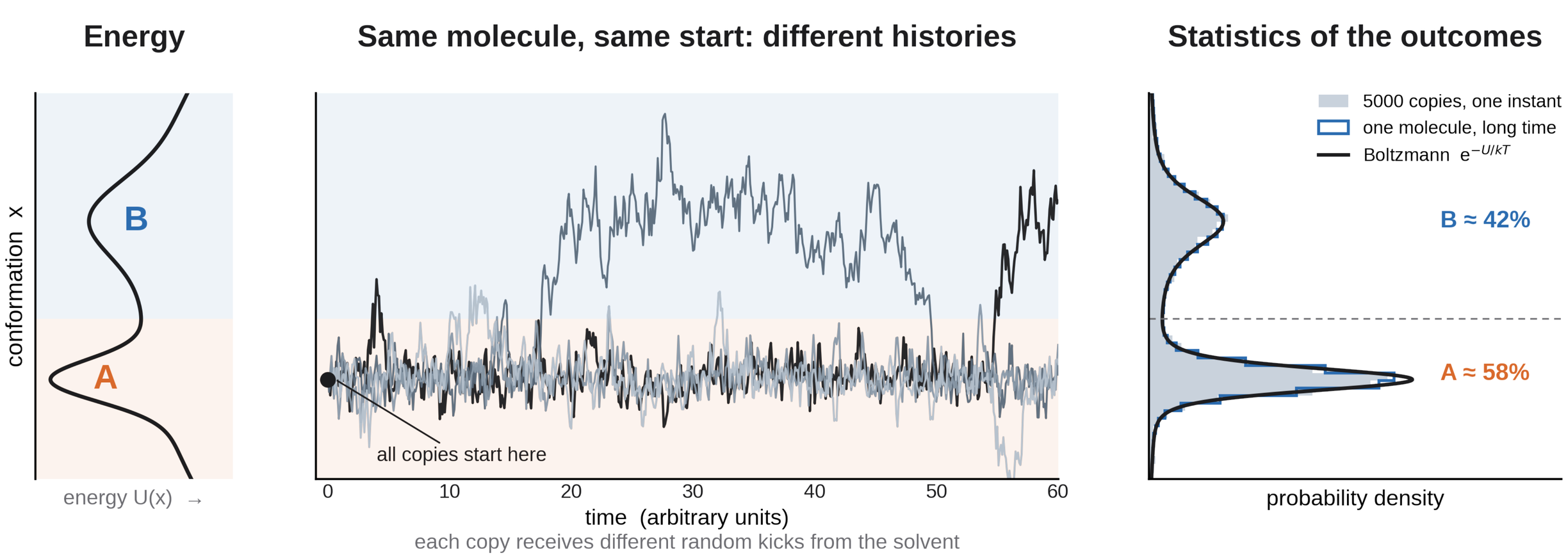
Statistical Mechanics: Ergodicity



1. **A single molecule (at constant temperature) will, given enough time, visit all the accessible conformations at random due to the collisions with the solvent. Each conformation is populated according to its energy and the temperature (the interaction with environment).**
2. **If at a given time one stops the movie of many many copies of the same system in the same condition the picture taken will display conformations according to the same statistics.**

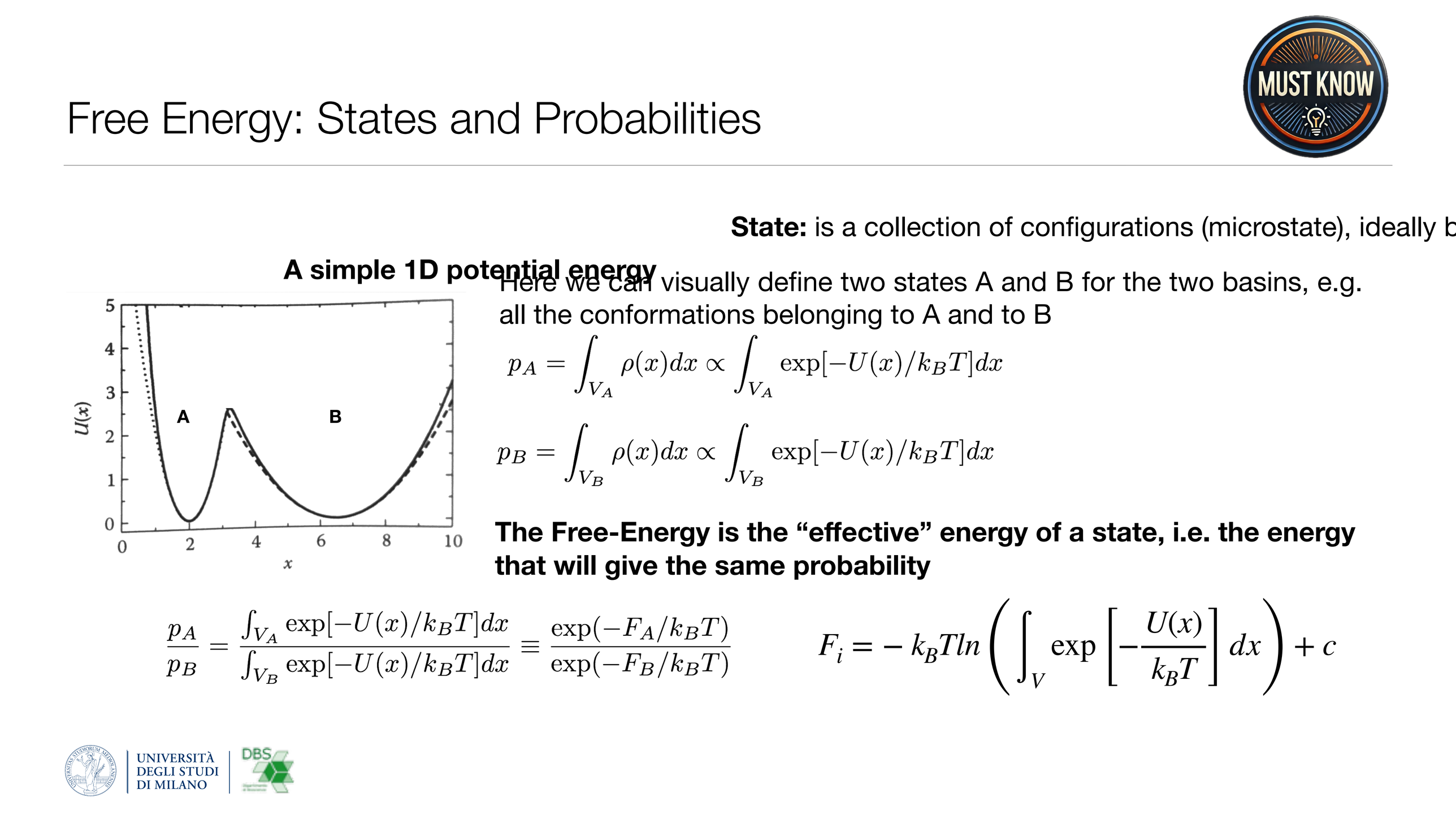
The collection of conformations in both cases (that is the sampling) is usually called an ensemble of conformations. This ensemble is characterised first of all by the conditions in which it has been acquired (constant temperature, pressure or volume, constant number of particles, etc.)

Statistical Mechanics: Ergodicity



Identical molecules starting from the same conformation follow different histories. The statistics of the outcomes are the same whether you watch many copies at one instant or one molecule for a long time, and they match Boltzmann.

Free Energy: States and Probabilities



State: is a collection of configurations (microstate), ideally b

A simple 1D potential energy

Here we can visually define two states A and B for the two basins, e.g. all the conformations belonging to A and to B

$$p_A = \int_{V_A} \rho(x)dx \propto \int_{V_A} \exp[-U(x)/k_BT]dx$$

$$p_B = \int_{V_B} \rho(x)dx \propto \int_{V_B} \exp[-U(x)/k_BT]dx$$

The Free-Energy is the "effective" energy of a state, i.e. the energy that will give the same probability

$$\frac{p_A}{p_B} = \frac{\int_{V_A} \exp[-U(x)/k_BT]dx}{\int_{V_B} \exp[-U(x)/k_BT]dx} \equiv \frac{\exp(-F_A/k_BT)}{\exp(-F_B/k_BT)}$$

$$F_i = -k_BTln\left(\int_V \exp\left[-\frac{U(x)}{k_BT}\right]dx\right) + c$$

Summary: relative probabilities, energy and time



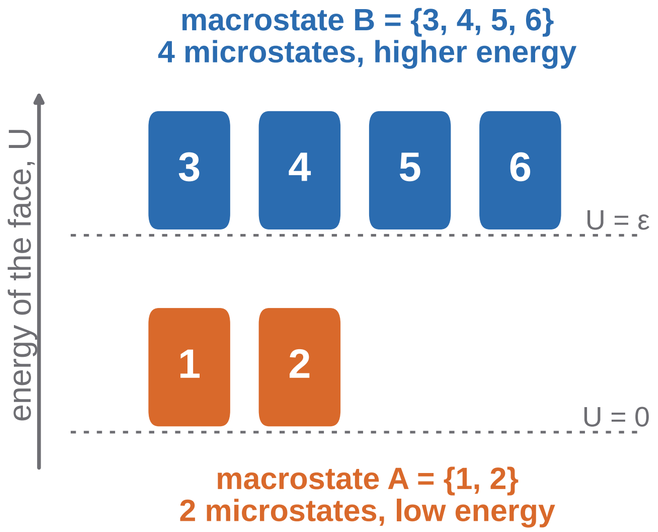
Microscopic processes are guided by **energy**, between two microscopic states their relative probability is associated exponentially to their energy difference.

Macroscopic processes (made of many microscopic ones) are guided by **free-energy**, between two macroscopic states their relative probability is associated exponentially to their free-energy difference, the (logarithm of the) number of microscopic states in a macroscopic state is its entropy, $S = k \ln W$.

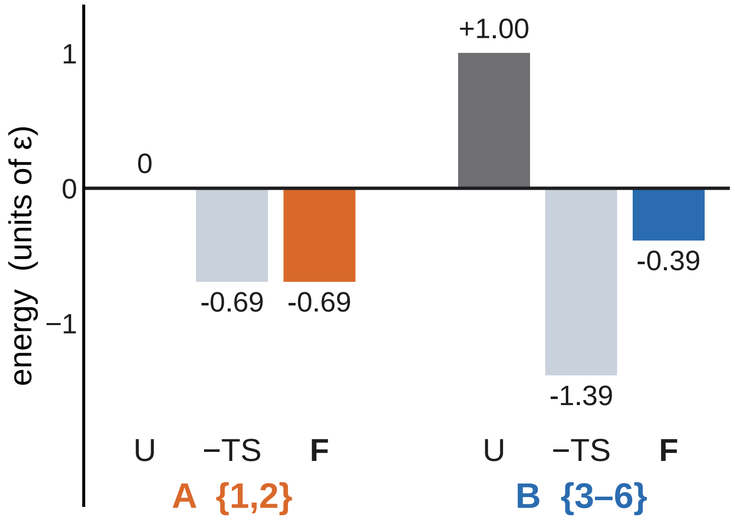
Macroscopic Processes happen in a given average time that is associated with the free-energy barrier separating them

Enthalpy and entropy: the case of the biased die

Biased die: energy levels

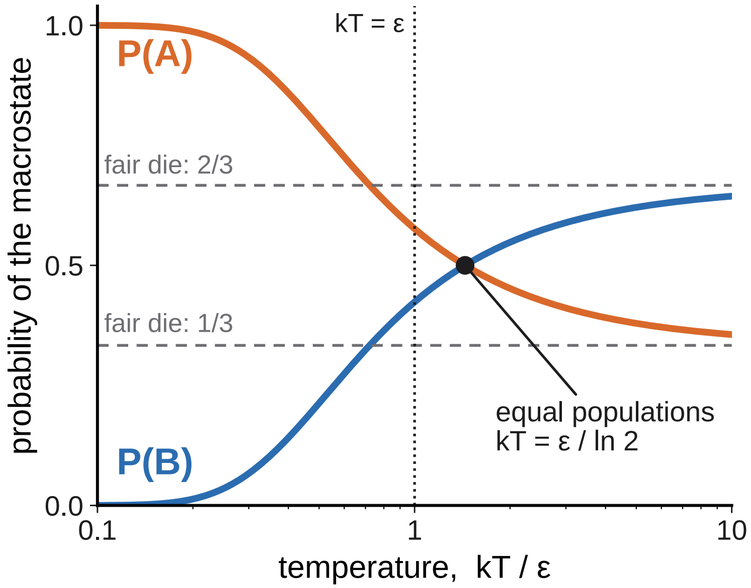


Free energy of a macrostate, kT = ε



F = U − TS: energy favours A, entropy favours B

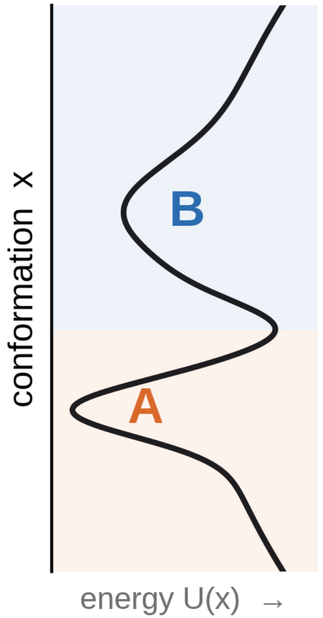
Populations vs temperature



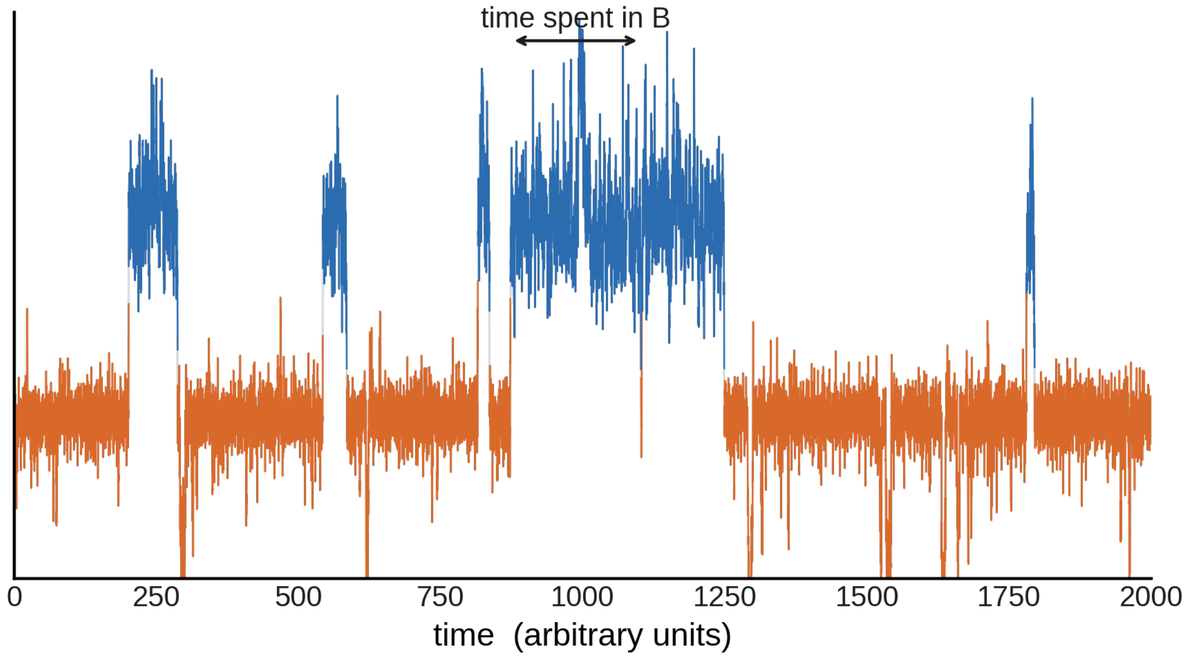
low T: energy wins (A) · high T: entropy wins (B)

A rate is a probability per unit time

Energy

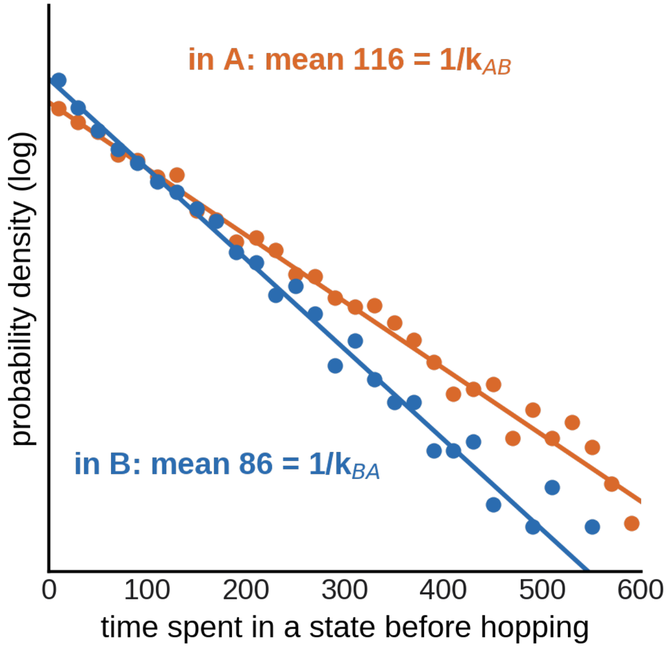


One molecule: random hops between A and B



12 transitions in this stretch · when the next hop happens is random

Waiting times are exponential

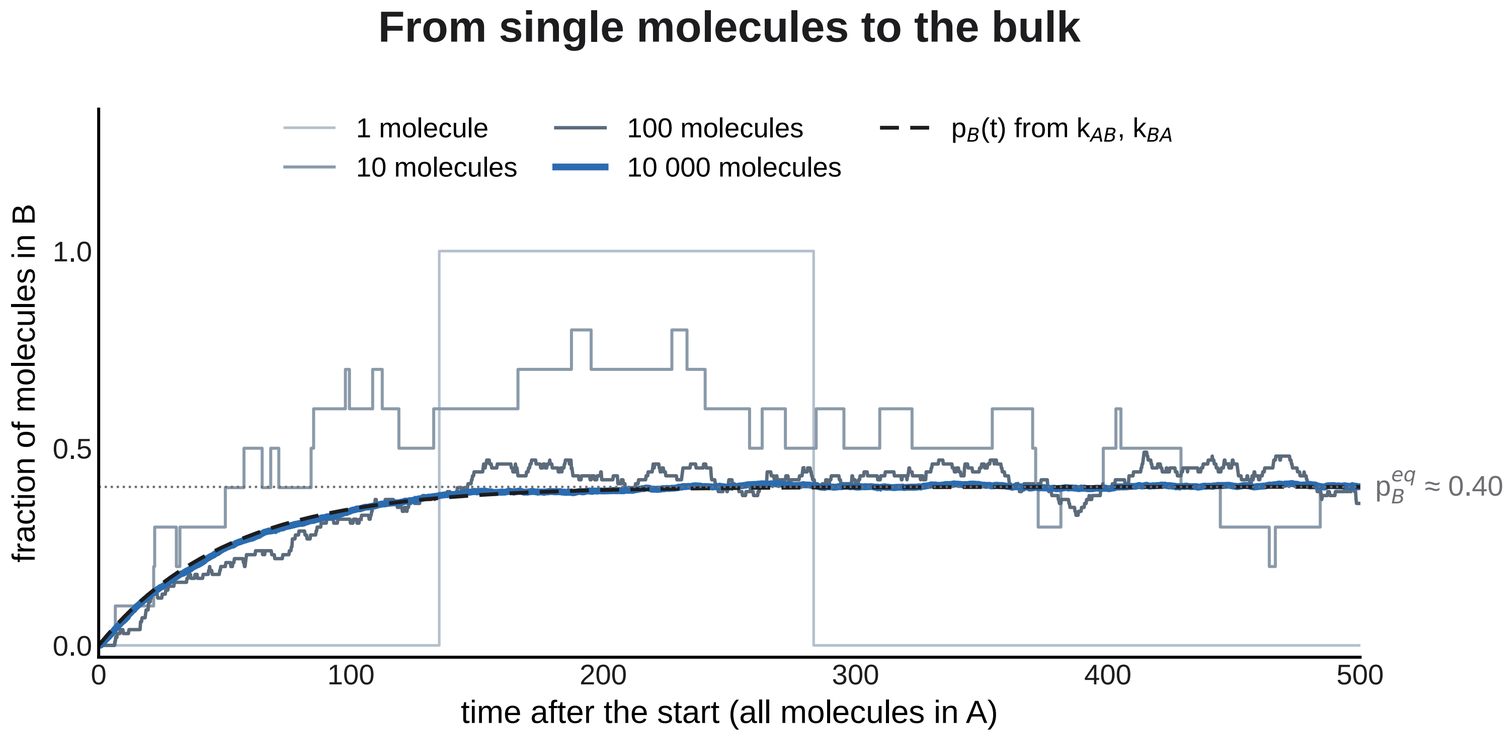


dots: simulation · lines: k·exp(−k t)

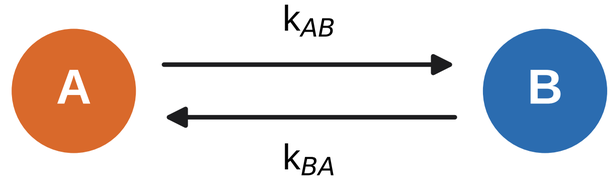
- A single molecule hops between A and B at random times
- The rate k_AB is the probability per unit time of leaving A: in a short dt the jump happens with probability k_AB·dt
- So the time spent in A is exponentially distributed, with mean 1/k_AB (like waiting for a 6 when rolling a die)

From single molecules to the bulk

- Start with all molecules in A and count how many are in B over time
- One molecule: a random jump. Many molecules: a smooth exponential



single molecules jump at random; their average relaxes smoothly and exponentially



$$dp_B/dt = k_{AB}\,p_A - k_{BA}\,p_B$$

change = flux in − flux out

$$p_B(t) = p_B^{eq}\,(1 - e^{-k_{obs}t})$$

single-exponential relaxation

$$k_{obs} = k_{AB} + k_{BA}$$

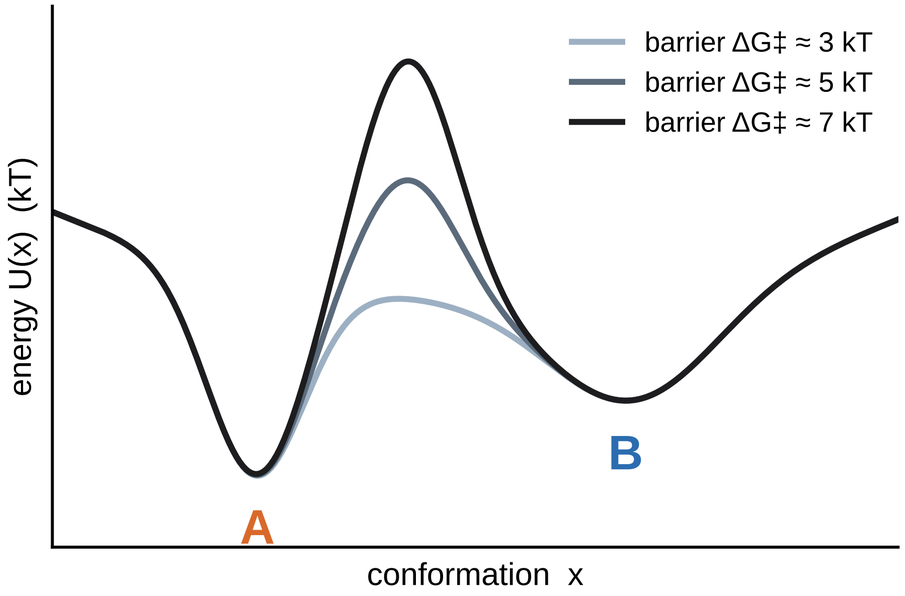
here ≈ 0.020: both directions count

$$k_{AB}\,p_A^{eq} = k_{BA}\,p_B^{eq}$$

equilibrium: the two fluxes balance

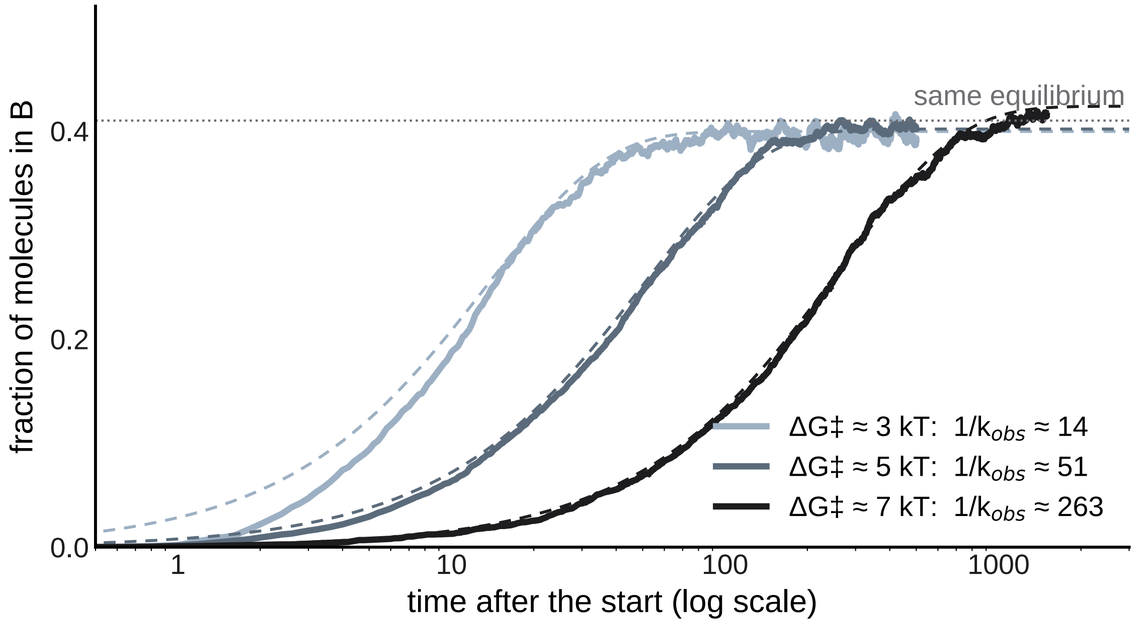
The barrier sets the speed, the free-energy difference sets the populations

Same basins, different barriers



basins A and B unchanged: same populations

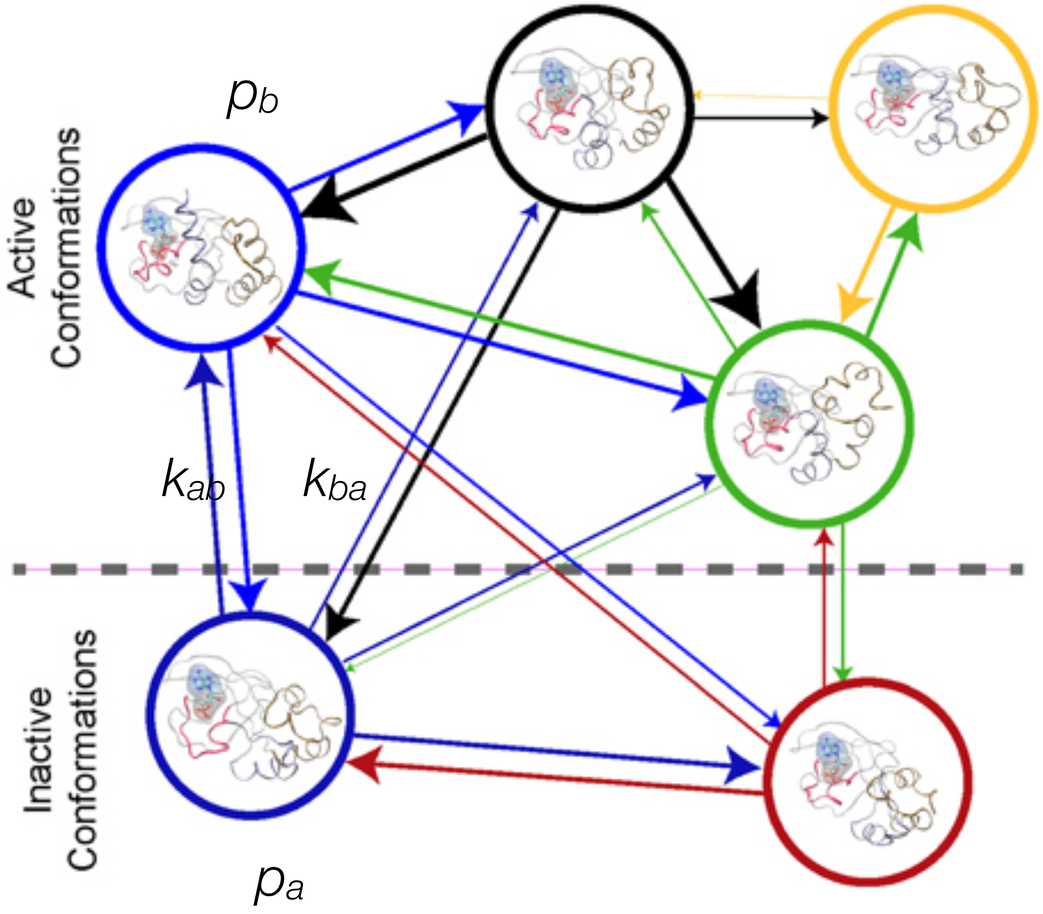
The barrier sets the speed, not the populations



thick: 5000–10 000 simulated molecules · dashed: $k_{AB}/k_{obs}\cdot(1 - e^{-k_{obs}t})$

- k ≈ k0·exp(–ΔG‡/kT): higher barriers make the crossing exponentially rarer
- Same basins, different barriers → same equilibrium, very different times to reach it
- Populations ↔ ΔG between states; rates ↔ ΔG‡ of the barrier

Summary: A stochastic picture of molecules in solution



Macrostate populations are associated with free energy differences:

$$p_i \propto \exp\left(-\frac{G_i}{RT}\right)$$

Exchange rates are associated with free energy barriers:

$$k_{ij} = k_0 \exp\left(-\frac{G_{ij}^{\ddagger}}{RT}\right)$$

Energy for these processes is provided by thermal energy, so essentially by collision with water molecules

An equilibrium bulk experiment ideally captures the contribution from all the states:

$$< O >= \sum_i p_i O_i = \frac{\sum_i O_i \exp\left(\frac{-G_i}{RT}\right)}{\sum_i \exp\left(\frac{-G_i}{RT}\right)}$$

Microscopically:

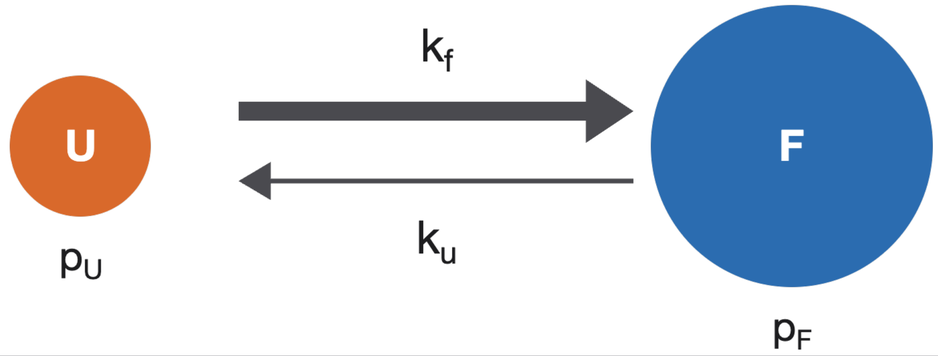
$$G_i = -RT \ln \int \exp\left(-\frac{U(r)}{RT}\right) \delta(i - i(r)) dr$$

Probability - Energy - Rates in Protein Folding aind Ligand Binding

Unfolded · Unbound **Folded · Bound**

Protein folding

U ↔ F

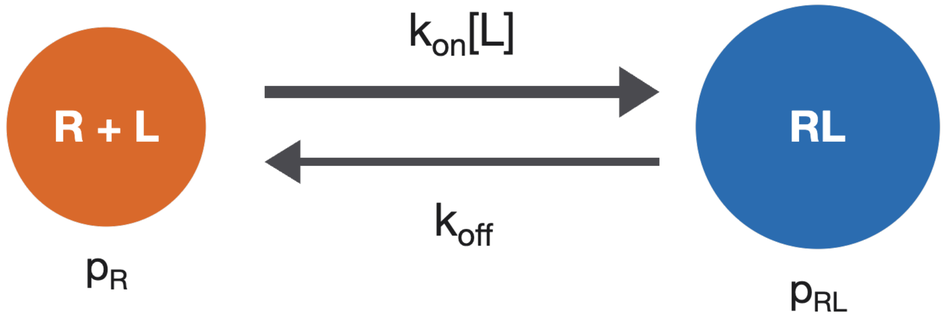


Equilibrium

$p_F / p_U = k_f / k_u$

Ligand binding

R + L ↔ RL



Equilibrium

$p_{RL} / p_R = k_{on}[L] / k_{off}$
$= [L] / K_D$

Populations ↔ free-energy difference

Rates ↔ free-energy barrier

node size shows population · arrow width shows rate (illustrative)

What do different experiments measure?

	Bulk thermodynamics (equilibrium spectroscopy, DSC, ITC)	Bulk kinetics (stopped-flow, T-jump, SPR)	Single molecule (smFRET, optical tweezers)
Populations, ΔG	✓	✓ via K = k_u/k_f, K_D = k_off/k_on	✓ histogram of states
ΔH, ΔS	✓ (calorimetry, or temperature dependence)	✓ from the temperature dependence of the rates	✓ from temperature or force dependence
Rates	✗	✓	✓ dwell times
Intermediates, heterogeneity, rare events	✗ averaged out	partly (extra exponentials)	✓

A stochastic picture: ligand binding

$$R + L \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} RL$$

If we start from a solution of free [R] and [L] we start forming RL with rate k_{on}, once we have some [RL] this can also dissociate with rate k_{off}, after some time we will be at equilibrium that is that the concentration of [R], [L] and [RL] will not change anymore.

$$K_d \equiv \frac{k_{off}}{k_{on}} = \frac{[R][L]}{[RL]} = C^0 \frac{p_{unbound}}{p_{bound}}$$

The **dissociation constant** can then be defined as the ratio between k_{off} and k_{on}, it is measured in (M), where lower values means stronger binding (i.e. lower concentrations of ligand are enough to bind)

$$C^0 = 1\ M$$

$$\Delta G_{\text{binding}} = k_{\text{B}} T \ln\left(\frac{K_{\text{d}}}{C^0}\right) = k_{\text{B}} T \ln\left(\frac{k_{\text{off}}}{k_{\text{on}} C^0}\right)$$

$$= k_B T \ln\left(\frac{p_{unbound}}{p_{bound}}\right)$$

This is the binding free-energy that is the amount of work that can be extracted or that must be provided for the binding reaction to happen, and you see is related to the rates as well as to the probabilities

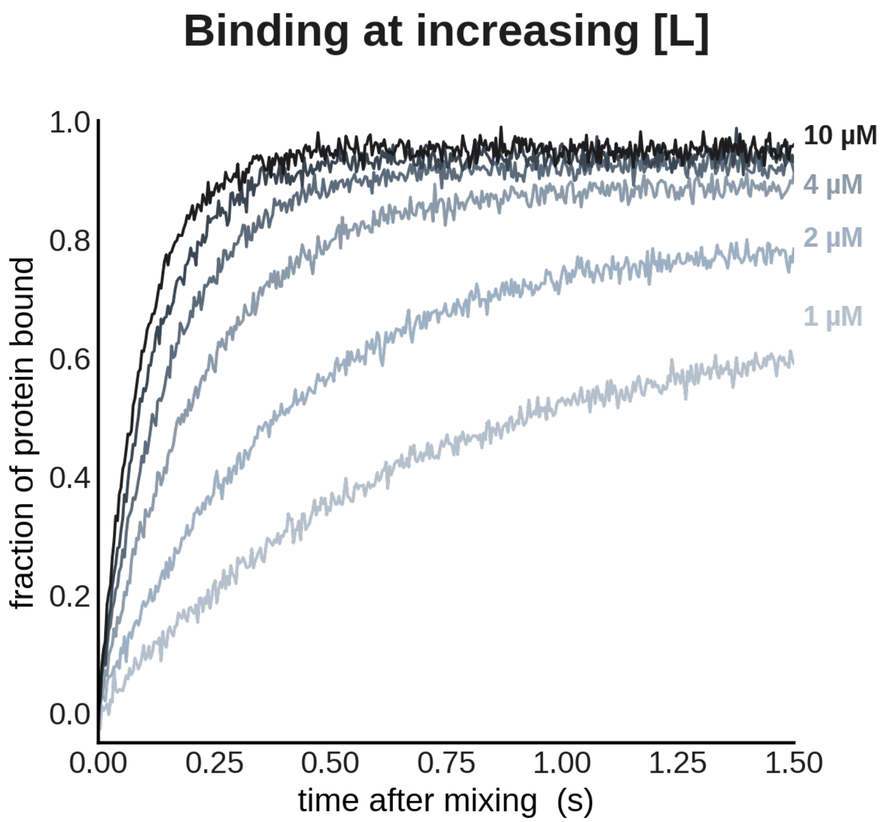
Ligand binding: a molecular example

$$R + L \underset{k_{off}}{\overset{k_{on}}{\rightleftharpoons}} RL$$

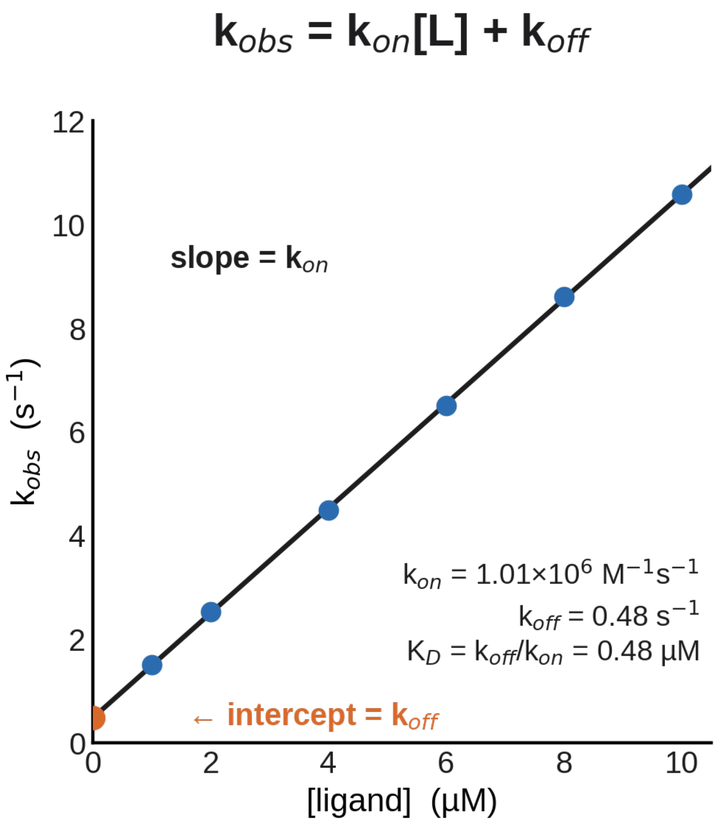
The k_{off} rate is the frequency of unbinding (s^{-1}) so that $k_{off}[RL]$ is the number of unbinding events per second per unit volume ($M\ s^{-1}$).

The k_{on} represent instead two processes (1) collision and (2) complexation. So this is the collision frequency times the complexation probability, this means that binding depends on concentration, indeed k_{on} is measured in ($s^{-1}\ M^{-1}$). $k_{on}[R][L]$ is the number of binding events per second, while $k_{on}[L]$ is the frequency of binding for a given concentration of ligand (excess of ligand).

Measuring binding kinetics



stopped-flow, ligand in excess



kinetics → thermodynamics: $\Delta G° = RT \ln K_D \approx$ -36 kJ/mol

- Mix protein with excess ligand and follow binding in time: a single exponential with k_obs = k_on[L] + k_off
- Repeat at several [L]: the slope is k_on, the intercept is k_off
- Kinetics → thermodynamics: K_D = k_off/k_on, ΔG° = RT ln(K_D/C°)

A stochastic picture: protein folding (same framework as ligand binding)

$$U \underset{k_u}{\overset{k_f}{\rightleftharpoons}} F$$

The k_u rate is the frequency of unfolding (s^{-1}) so that $k_u[F]$ is the number of unfolding events per second.

The k_f rate is the frequency of folding (s^{-1}) so that $k_f[U]$ is the number of folding events per second.

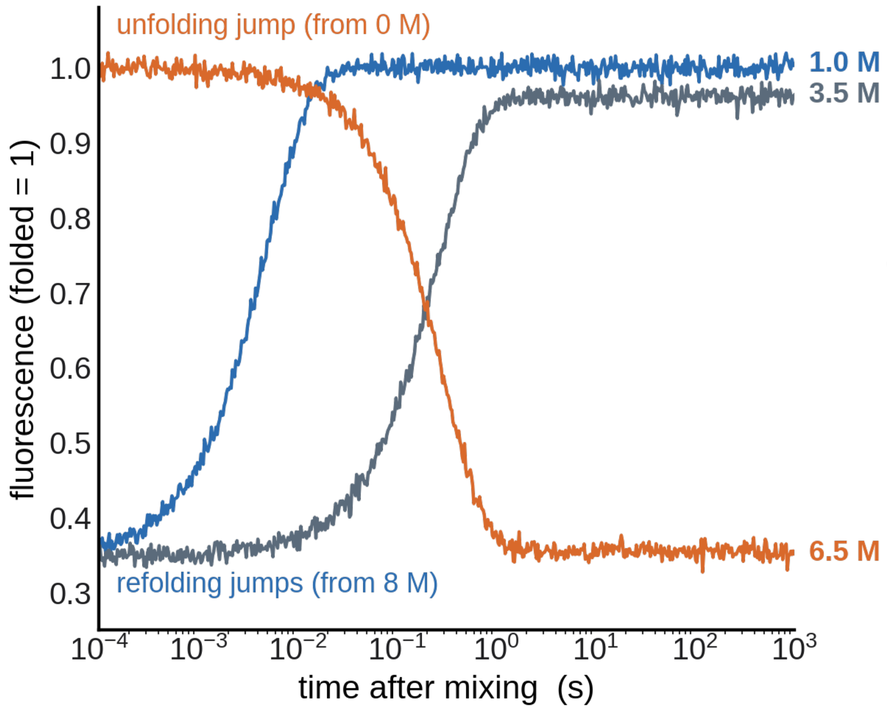
$$K = \frac{k_U}{k_F}$$

The two rates are measured by stopped-flow after a jump in denaturant concentration (next slide).

$$\Delta G_F = RT \ln K = G_F - G_U$$

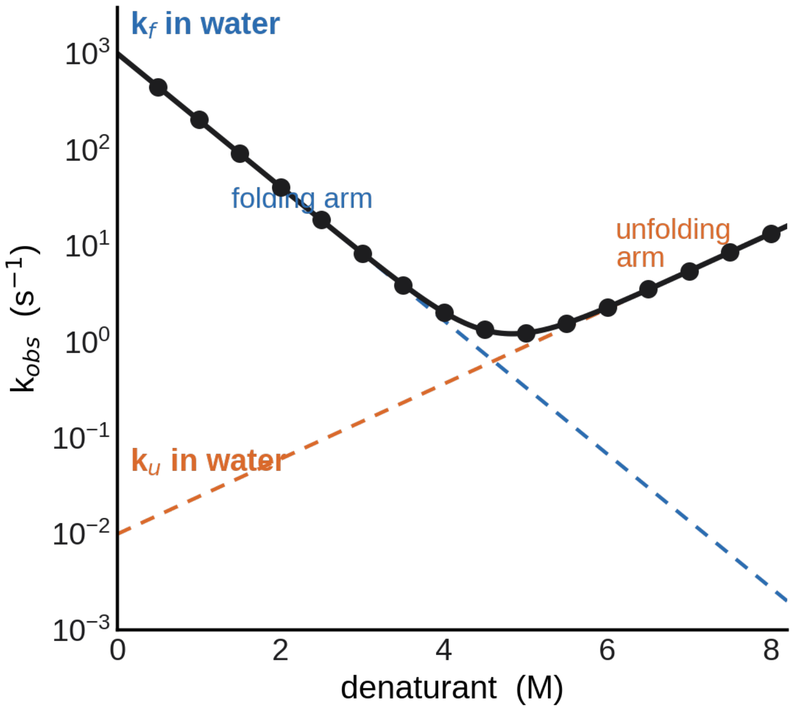
Measuring folding kinetics: the chevron plot

Stopped-flow relaxations



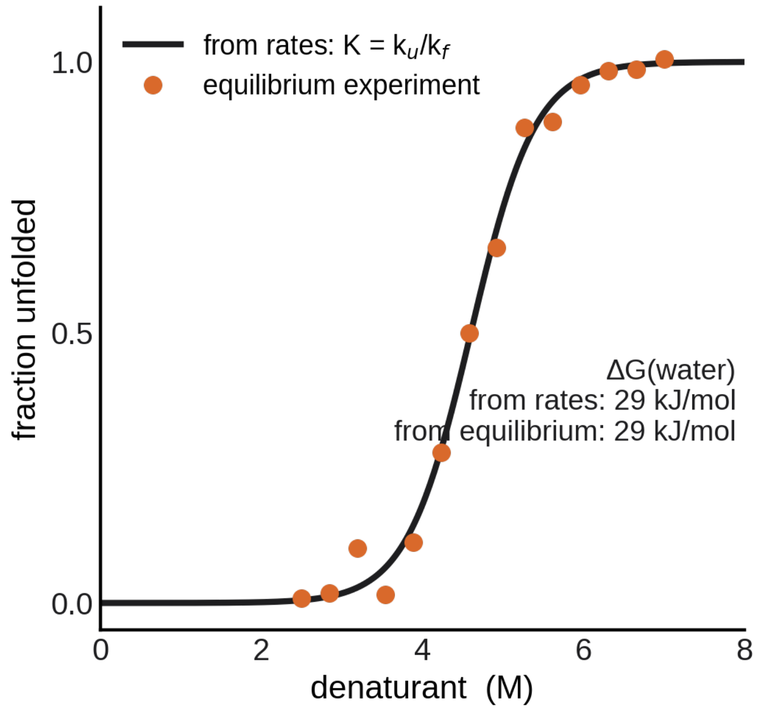
single exponentials, $k_{obs} = k_f + k_u$

Chevron plot



extrapolate both arms to 0 M

Kinetics gives stability



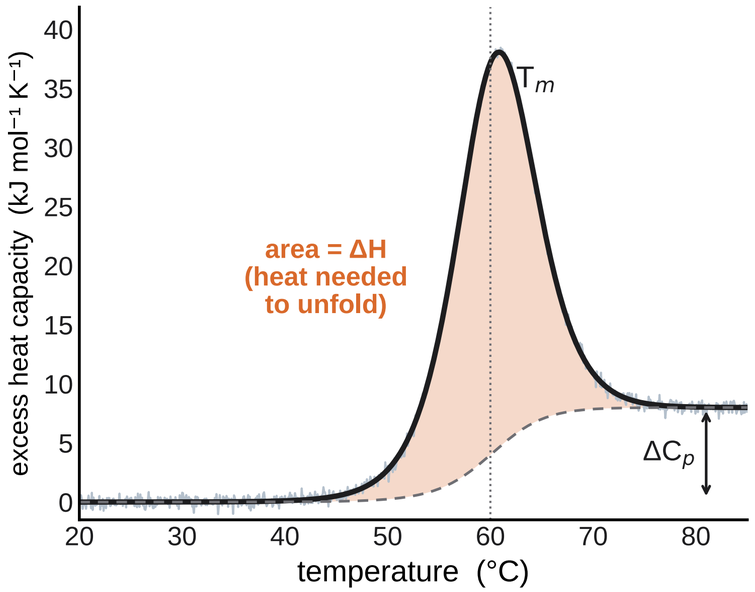
agreement = test of the two-state model

- Rapidly mix the protein into a new denaturant concentration; fluorescence relaxes with k_obs = k_f + k_u
- k_obs vs [denaturant] is V-shaped (the "chevron"): k_f dominates at low denaturant, k_u at high denaturant
- Extrapolating the two arms to 0 M gives k_f and k_u in water, and therefore K and the stability ΔG_unfolding = RT ln(k_f/k_u)

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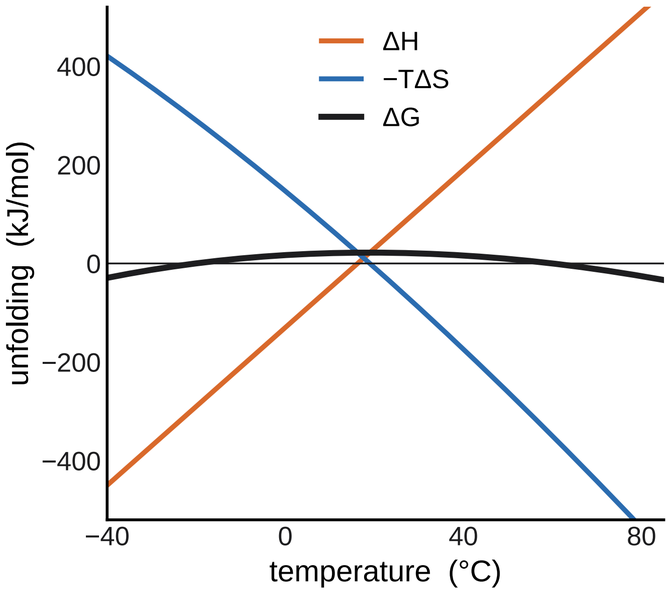
Calorimetry (DSC): heat reveals stability, enthalpy and entropy

DSC thermogram



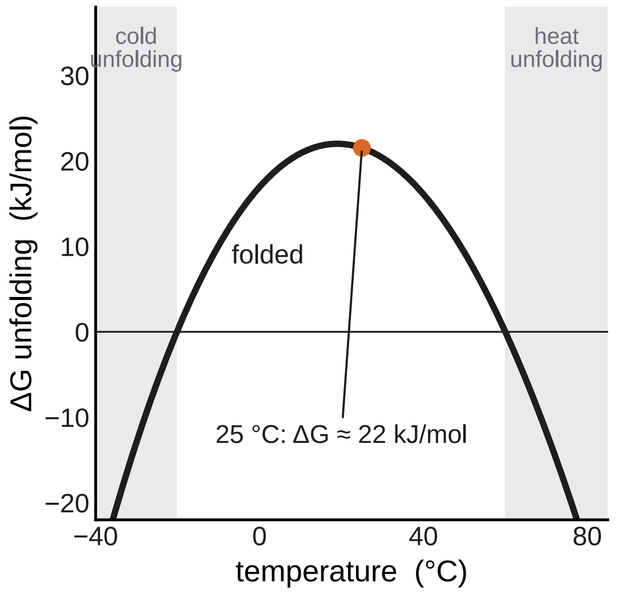
grey: simulated data · black: two-state model

Large opposing terms…



ΔH and TΔS both grow with T ($\Delta C_p > 0$)

…small stability



stability curve, ΔG = 0 at T_m

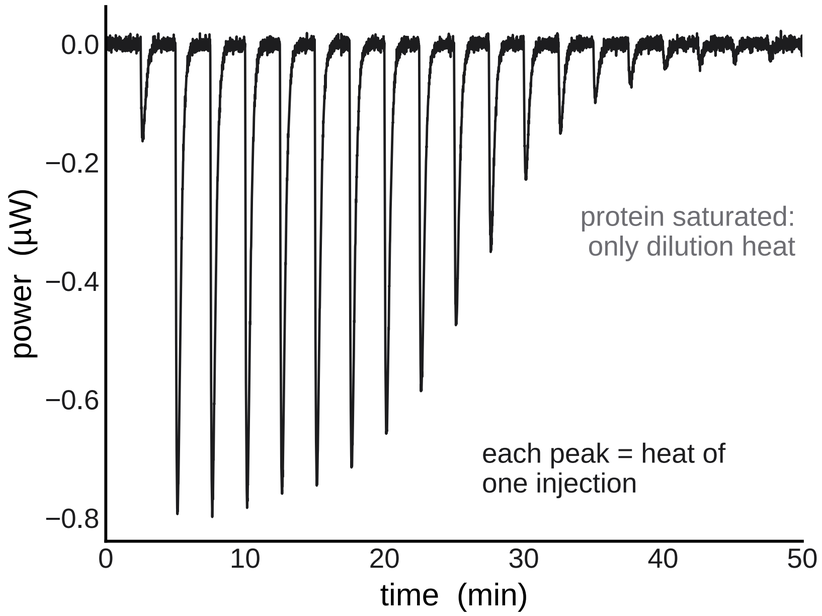
- Differential scanning calorimetry: heat the protein slowly (~1 °C/min) and measure the extra heat it absorbs compared with buffer
- Peak at T_m (half unfolded); area = ΔH of unfolding; baseline step = ΔC_p
- From T_m, ΔH and ΔC_p we get ΔG(T) at any temperature: the stability is a small difference between large ΔH and TΔS

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Calorimetry gives ΔG, ΔH and ΔS, but no rates: the same thermogram can come from a protein that folds in microseconds or in minutes.

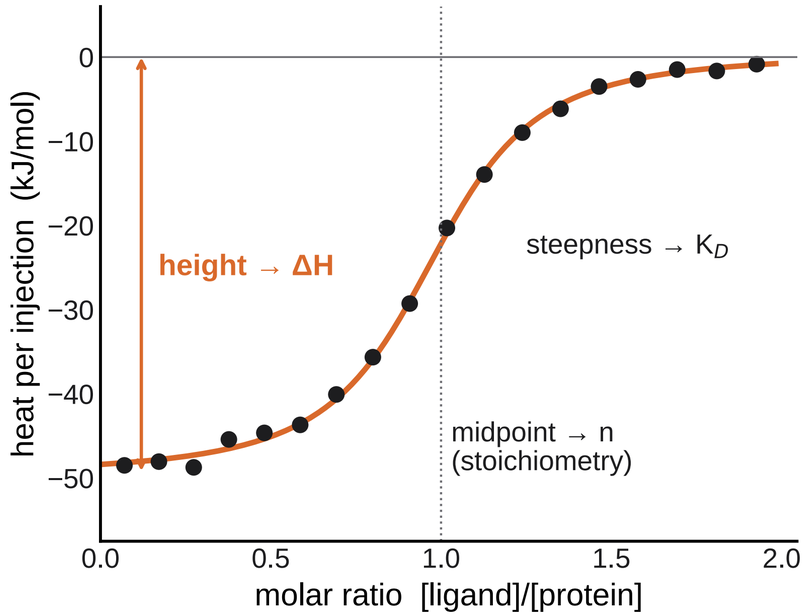
Calorimetry (ITC): the full thermodynamics of binding in one experiment

ITC: titrating ligand into protein



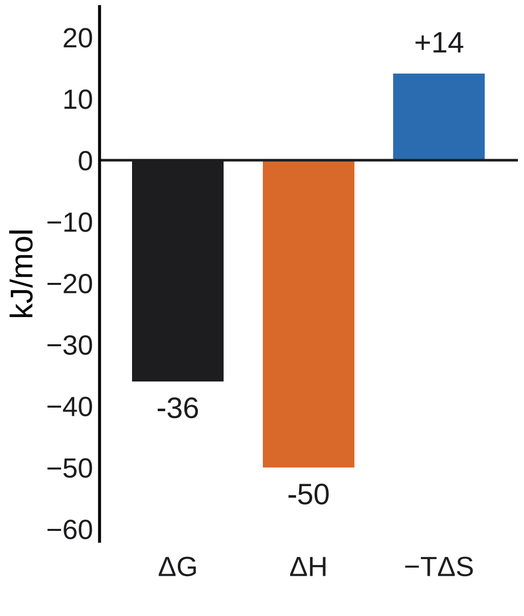
exothermic binding: downward peaks

Binding isotherm



dots: integrated peaks · line: 1:1 model, c = n[P]/K_D = 40

Signature



K_D = 0.5 µM, 25 °C

- Isothermal titration calorimetry: inject ligand into the protein in small steps at constant T and measure the heat of each injection
- The binding isotherm gives K_D (steepness), ΔH (height) and n (midpoint)
- Then ΔG = RT ln(K_D/C°) and −TΔS = ΔG − ΔH: binding can be driven by enthalpy, entropy, or both

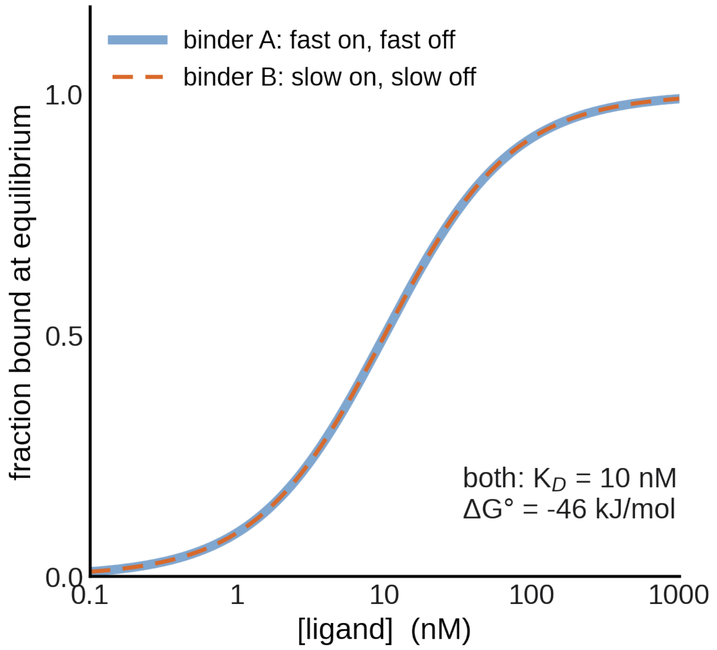
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DBS

ITC gives K_D, ΔH and n, but not k_on and k_off: many fast or slow pairs of rates give the same K_D

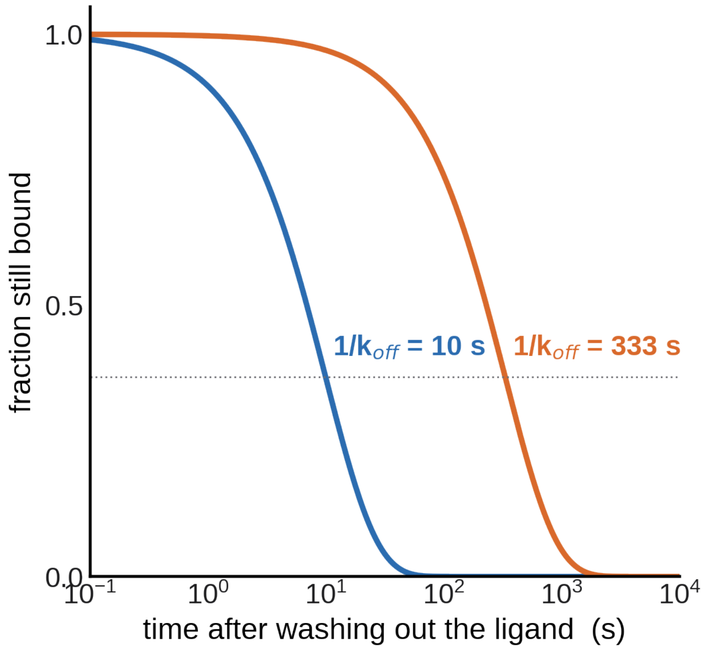
Thermodynamics does not give kinetics

Same thermodynamics…



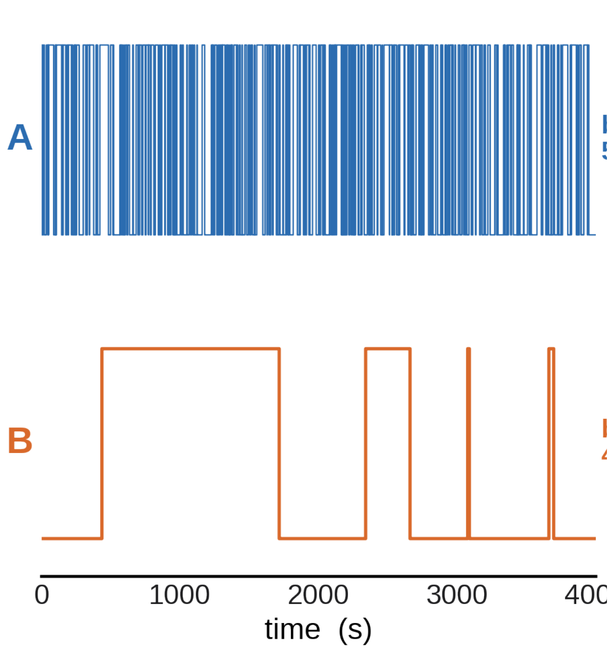
titration, ITC: identical — no information on rates

…different kinetics



bulk kinetics: k_{on}, k_{off} and also $K_D = k_{off}/k_{on}$

Single molecules: both

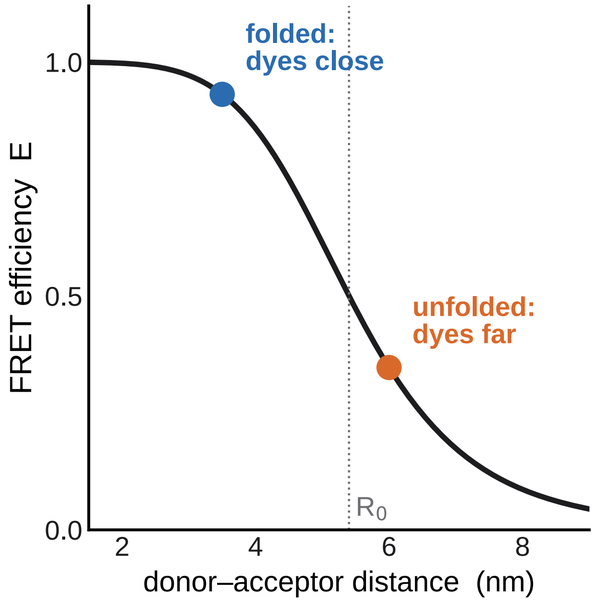


[L] = K_D: occupancy → K_D, dwell times → k_c

- Two binders with the same K_D = 10 nM (ΔG° ≈ –46 kJ/mol): B is 30 times slower than A
- Titration or ITC: identical curves; no equilibrium measurement can tell them apart
- Bulk kinetics (washout): residence times of 10 s vs 333 s
- One molecule observed at [L] = K_D: both are bound about half of the time, but A switches faster

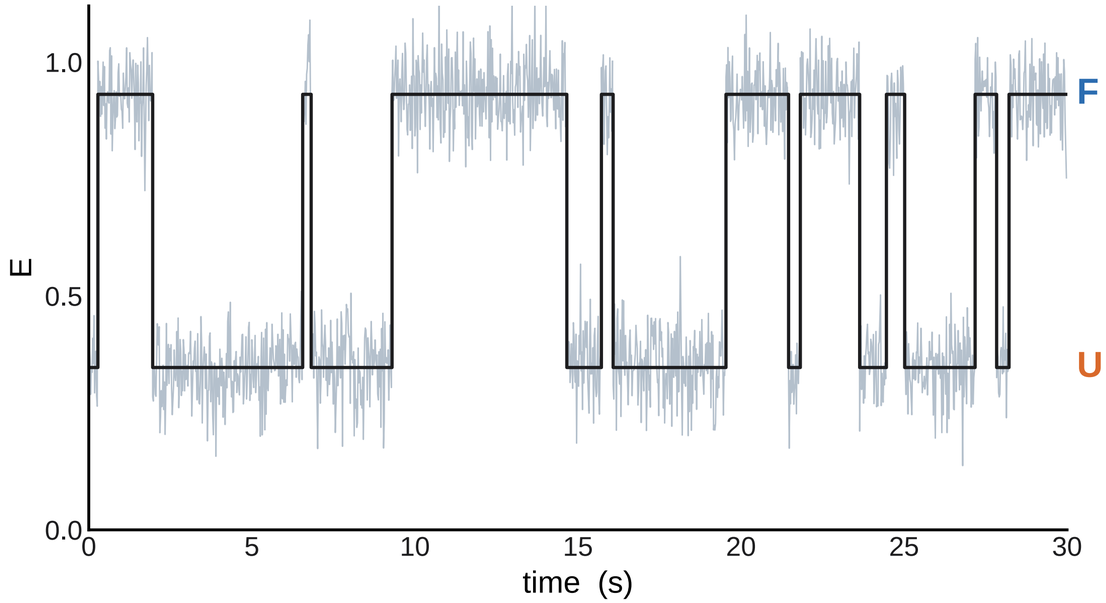
Single molecules (smFRET): watching one protein fold and unfold

A molecular ruler



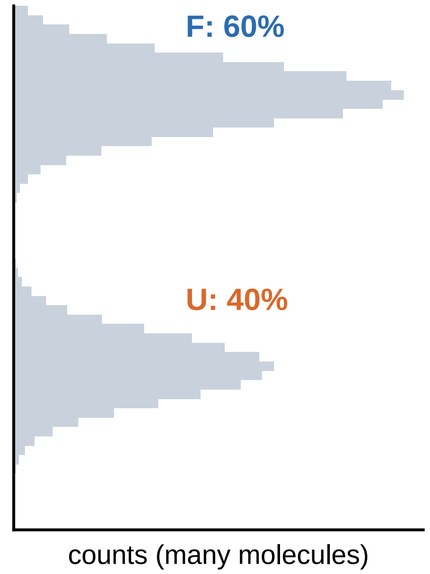
$E = 1 / [1 + (r/R_0)^6]$

One protein folding and unfolding



grey: simulated signal (20 ms frames) · black: assigned state → dwell times → k_f, k_u

Populations

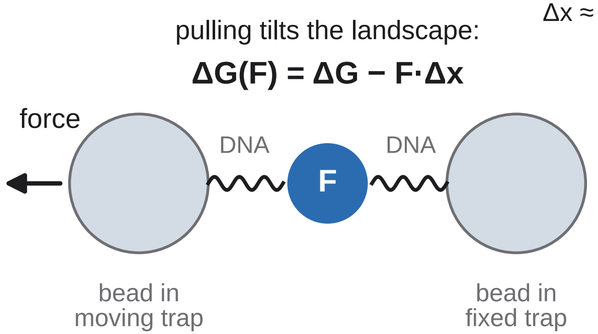


$\Delta G = -RT \ln(p_F/p_U) \approx -1.0$ kJ/mol

- Two dyes on the protein: FRET efficiency E reports their distance, between about 2 and 8 nm
- One molecule switches between folded (high E) and unfolded (low E) at random times
- Histogram → populations → ΔG; dwell times → k_f and k_u; also intermediates and heterogeneity hidden in the bulk

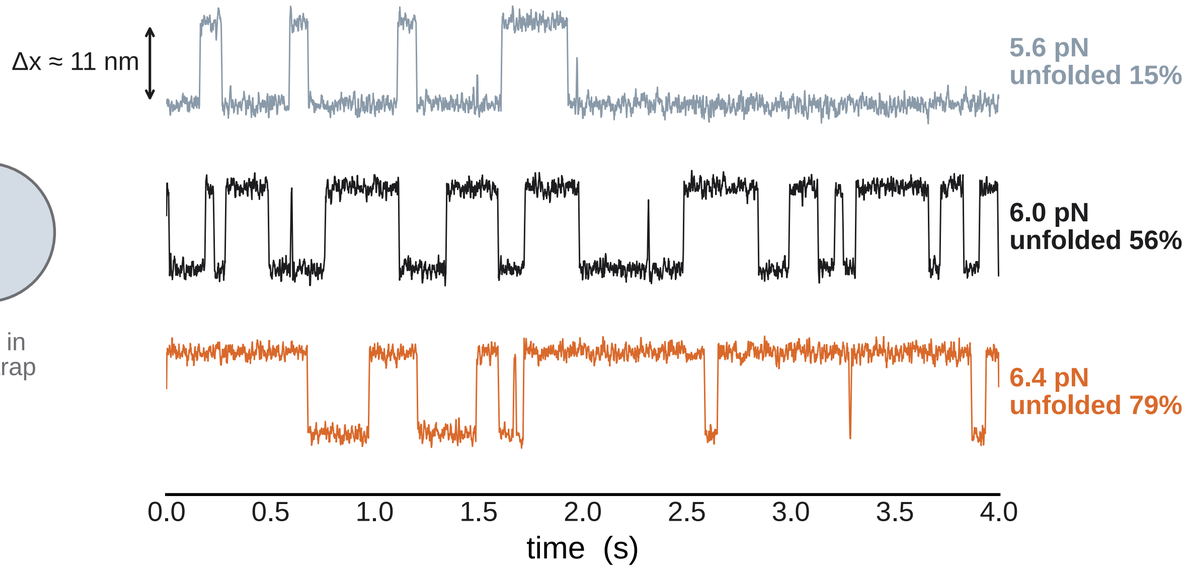
Single molecules (optical tweezers): force as a control parameter

Optical tweezers



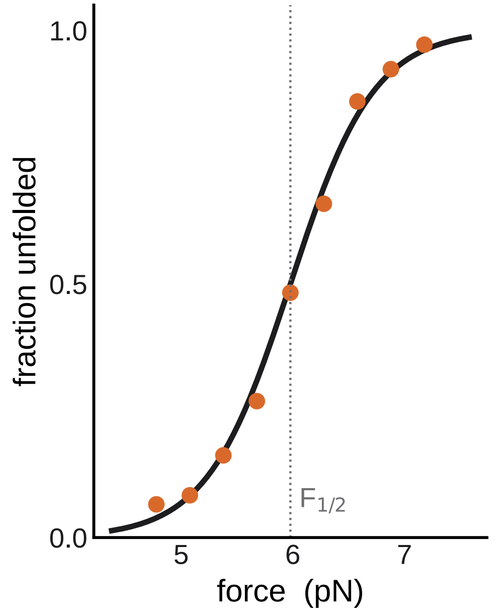
~0.1 pN and ~1 nm resolution

Hopping at constant force



simulated extension traces: up = unfolded

Force titration



ΔG = $F_{1/2}$·Δx

- The protein is tethered between two beads held in laser traps; we control the force and measure the extension
- A force tilts the landscape: ΔG(F) = ΔG – F·Δx, so the unfolded state becomes more populated as F increases
- At F_1/2 folded and unfolded are equally populated: ΔG ≈ F_1/2·Δx; rates change exponentially with force

Recap

- Observing a molecule is a random draw: thermal kicks make molecular outcomes stochastic
- Microscopic probability ↔ energy: $p \propto \exp(-E/kT)$
- Macroscopic (free-energy) probability ↔ entropy and energy of a state
- A rate is a probability per unit time: single molecules hop at random, the bulk relaxes exponentially
- Populations ↔ free-energy differences; rates ↔ free-energy barriers
- Ligand binding and protein folding: same framework, two different molecular events
- Bulk kinetics gives the rates and, through K = k_forward/k_backward, also the thermodynamics
- Bulk thermodynamics (equilibrium curves, calorimetry) gives ΔG, ΔH and ΔS, but no rates
- Single-molecule experiments give both, plus what averages hide

The common theme of Structural Bioinformatics: sampling and energy (or score, or loss function)



In the next lectures we will see how the problem of SAMPLING and the problem of the ENERGY or SCORING FUNCTION or LOSS FUNCTION, are interconnected and always present. These are the problems to cope with when doing MD simulation, structure prediction, docking, protein design, …

You can think of the next lectures as specific strategies to cope with these two problems when trying to address specific problems. All the algorithms we will see are related to them (MD, MC, EM, ML/AI, …)

Questions:

- How do you calculate the average outcome of a stochastic process?
- What is the difference between the standard deviation and the standard error of the mean?
- Which is the probability distribution function of the average of a process?
- What is the difference between the absolute probability of an event and the relative probability of that event with respect to another?
- What is the relationship between the probability of a macrostate and its free-energy?
- What is the relationship between the probability of a configuration and its energy?
- What is the free energy change of a process?
- …